Michelson Interferometry Measurements Experimental Outline

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Abstract

Michelson interferometry is a popular interferometer arrangement used to make precise measurements with a laser beam. We use Michelson interferometry to measure three quantities: (1) the wavelength of a HeNe laser beam, (2) the distance between wavelengths of the yellow Mercury doublet, and (3) the index of refraction of a transparent material. We use a table-top interferometer connected to cameras to make these measurements.

1 Introduction

Interferometers are powerful instruments which harness the wave-like behavior of light to make precise measurements by analyzing a split-beam's interference pattern. Michelson interferometry is a popular arrangement of such tools, becoming famous when it first proved that light did not propagate through a "luminiferous ether" as previously believed [1]. The apparatus uses one movable and one stationary mirror to create an interference pattern of a split light source where the two beams travel paths of different lengths [2]. Moving the mirror and analyzing the resulting pattern provides for precise measurements of the wavelength of the light, as first studied, or other measurements as we examine here.

Figure 1 shows a basic schematic of the Michelson interferometer. Mirror 1 (M1) and Mirror 2 (M2) are normal to each other and the centralized beam splitter (BS) is at a 45°angle to both. The BS refracts the light source to two distinct paths where the beams travel different lengths. M1 is stationary while M2 is free to move, allowing one to create different interference fringe patterns. The final pattern can be viewed on the screen at the top of the diagram. Note that the first beam (1) passes through the BS 3 times while beam (2) passes through only once.

We use this apparatus to make three different measurements. We first measure the wavelength of a HeNe laser beam and Thallium lamp by counting fringes manually with camera images and automatically using a photodetector. We then measure the separation in wavelength between the yellow Mercury doublet. Finally, we insert a thin piece of x material to measure its refractive index.

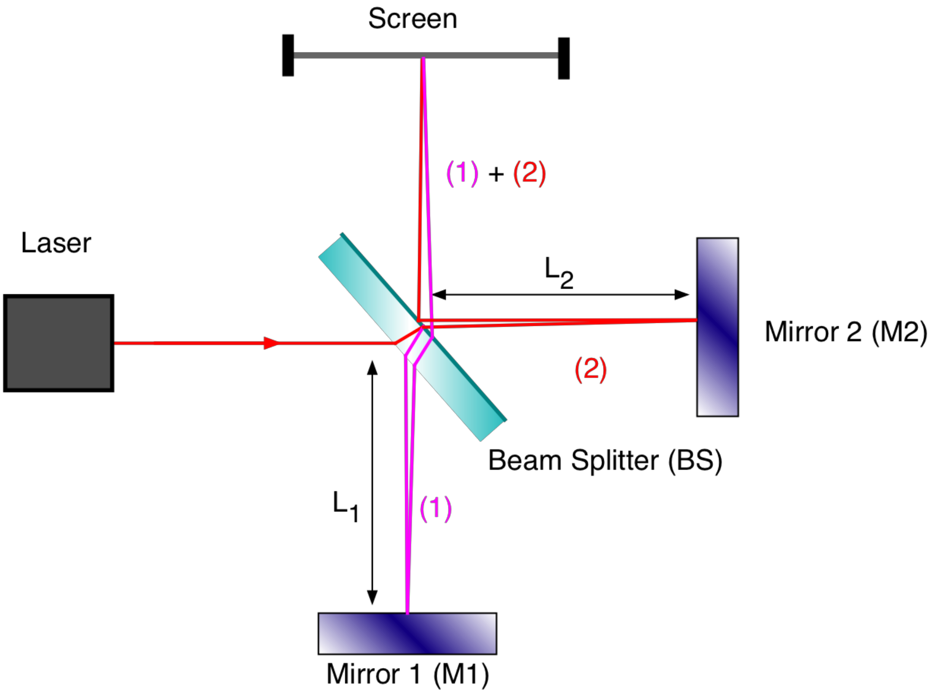

Figure 1: Schematic of a Michelson interferometer. The laser beam passes through the semi-transparent beam splitter (BS) such that one beam travels to Mirror 1 through the BS 3 times, while the other travels through only once to Mirror 2 and is then reflected to the screen. The two colors show the paths of the two beams. Image from wanda.fiu.edu/boeglinw.

2 Safety Considerations

Interferometer intrinsically implies the use of a laser. Before the start of the experiment, we will have completed the basic laser safety training. We will also remove reflective objects like jewelry and watches when in the lab. Finally, we will maintain awareness of the laser pointing at all times so as not to look directly through the beam. This will require keeping our head above the plane of the experimental apparatus at all times.

3 Measuring HeNe laser and Thallium Spectral Line Wavelengths

3.1 Theory

For the first measurement we record the wavelength of the HeNe light source used for the remainder of the experiment. This measurement is straightforward and follow from the geometry of the experimental apparatus shown in Figure 1. The beams can create three possible interference patterns: hyperbolic fringes (when the mirrors are not parallel and path lengths L1 and L2 are not equal), vertical line fringes (when the mirrors are tilted from normal to the plane of the experiment), and circular fringes which occur when the apparatus is arranged as shown in Figure 1.

Suppose that beam one travels a total distance d_1 and beam two travels d_2. Then the beams will constructively interfere and create a bright ring where $|d_1 - d_2| = n\lambda$ where n is an integer, and the destructive interference (dark rings) will occur when $|d_1 - d_2| = n'\frac{\lambda}{2}$, where n' is an odd integer [2]. From this, we know that the same pattern will repeat itself when the mirror M2 moves a distance $\lambda/2$. Measuring this distance in the experimental apparatus allows us to solve for the laser beam's wavelength λ.

To measure the Thallium spectral line, turn off the HeNe laser. Use the Thallium lamp and repeat the measurement.

3.2 Methods

3.2.1 Materials

The materials required for this part of the experiment are:

- Base for experimental apparatus
- 2 surface aluminum mirrors (one stationary; one moveable)
- Beam splitter
- HeNe laser
- Thorlabs model DCC1545M camera
- USB cord
- Laptop (data analysis and visualization)
- Thallium lamp
- White card or screen
- ThorLabs DET36A photodetector

3.2.2 Experimental Setup

Following [3] and [4], place one aluminum mirror on the side of the base board opposite the laser entrance. Place the other mirror on the movable base on an adjacent side so that its face is perpendicular to the other mirror. Place the beam splitter in the middle of the board at a 45°angle to both mirrors so that the light will be refracted as in Figure 1. Align the two entrance mirrors so that the laser beam will hit the center of the first mirror. Use the green laser for alignment, carefully moving the mirrors as necessary until the set up aligns with Figure 1. Make sure the laser stays in the same horizontal plane, using the entrance mirrors to adjust the incidence as necessary. Make sure not to bend down such that the laser could be at eye level; always stay above the plane of the set up looking down to make adjustments.

Plug in the camera to the USB port of the desktop if it is not already. Focus the camera on the screen where the interference fringes appear after the beam has been reflected. Record a video of the fringes for analysis.

Replace the camera with the photodector, making sure that the center of the fringe pattern aligns with the center of the chip. Activate the detector by connecting it a HP

5315B Universal Counter. Select "Totalize Start" then "DC" then "Atten x1", and choose "Level" for the trigger. Make sure to hit the "reset" button after each measurement.

3.3 Analysis and Interpretation

This section will include a table of our measurements for λ using the camera and using the photodetector. We will tkae several measurements will both instruments and include the average and standard deviation in our final estimate. We expect a value around 632.816 nm.

Uncertainties will arise from manually moving the movable mirror and estimating when the fringe pattern repeats itself by eye when using the camera.

4 Measuring Separation of Mg Doublet Lines

4.1 Theory

The Mercury spectrum has two lines in yellow/green light closely spaced, requiring special techniques to measure the difference in their wavelengths. As described in [4], two spectral lines with similar wavelengths will have overlapping interference patterns with this experimental set up, resulting in either high or low contrast rings (i.e. clearly distinguishable alternating bright and dark rings, or rings which appear more "washed out"). Thus for a given angle θ, the maximum or minimum, on the screen at length d away will appear at $2d\cos(\theta) = (m + 1/2)\lambda$ and $2d\cos(\theta) = m\lambda$, respectively.

If the spectral lines peak at wavelengths λ_1 and λ_2 then we can find their difference $\delta\lambda$ by $\lambda_2 = \lambda_1 + \delta\lambda$ and $\Delta\lambda/\lambda_1 \ll 1$. Suppose there is a dark ring at angle θ on the screen. Then we know that $2d\cos(\theta) = m_1\lambda_1 = m_2\lambda_2$ and that the next dark fringe appears at $2d'\cos(\theta) = m'_1\lambda_1 = m'_2\lambda_2$, where $m'_2 = m_1 + k$ and $m'_1 = m_1 + k + 1$ and k is an integer. This yields $2(d' - d)\cos(\theta) = (k + 1)\lambda_1 = k\lambda_2$. Substituting the expression for $\Delta\lambda$ yields:

$$\Delta\lambda = \frac{\lambda_1\lambda_2}{2\Delta d\cos(\theta)} \tag{1}$$

4.2 Methods

4.2.1 Materials

This experiment requires the same materials listed in Section 3.2.1, as well as the following materials:

- Mercury light source
- Yellow Hg lines filter

4.2.2 Experimental Set Up

- Set up the apparatus in the same configuration as shown in Fig. 1 and described in 3.2.2. Use the Mg light source instead of the laser.
- Use the camera and not the automatic photodetector.

4.2.3 Experimental Procedure

- Adjust the mirrors the circular fringe pattern appears on the screen/wall. Note the color will be mostly blue. Now place the yellow Mg filter in front of the light source.
- Focus the camera on the bullseye pattern and record the data by taking an image
- Record both the high contrast and low contrast cases

4.3 Analysis and Interpretation

Assuming $\cos(\theta) \approx 1$ because the fringes are very close to the center of the interference pattern, Equation 1 simplifies to $\Delta\lambda = \lambda_{\text{avg}}^2/2\Delta d$. We will use this equation and record several measurements of λ_{avg} and Δd to estimate its mean and uncertainty. We will include an image of the fringe pattern that we measure.

Additional sources of uncertainty come from our approximation of the Mercury line. We could use a yellow filter then a green filter for a potentially more accurate measurement.

5 Measuring Refraction Indices

5.1 Theory

Following the derivation in Fendley [5], suppose there is a plate with index of refraction n that we want to measure using a light source with wavelength λ. The beam that passes through the plate of thickness d at an angle ϕ_i to the mirror will be displaced by

$$\frac{2\pi n\overline{PQ}}{\lambda} = \frac{2\pi nd}{\lambda\cos(\phi_r)} \tag{2}$$

when going from P to Q and the other beam will be displaced by

$$\frac{2\pi\overline{PS}}{\lambda} = \frac{2\pi d\cos(\phi_i - \phi_r)}{\lambda\cos^2(\phi_r)} \tag{3}$$

such that the phase difference between the two beams becomes

$$\Delta = 2\frac{2\pi nd}{\lambda\cos(\phi_r)} - \frac{2\pi d\cos(\phi_i - \phi_r)}{\lambda\cos(\phi_r)} \tag{4}$$

where ϕ_r is the angle of refraction and $\sin(\phi_i) = n\cos(\phi_r)$.

If $\phi_i = 0$, then the displacement is $\Delta_o = 2(2\pi nd/\lambda - 2\pi d/\lambda)$. By changing the angle of incidence from $\phi_i = 0$ to ϕ_i, the resulting inference pattern of m fringes can be analyzed such that

$$m = \frac{2d}{\lambda}\left(n\left(\frac{1}{\cos(\phi_r)} - 1\right) + 1 - \frac{\cos(\phi_i - \phi_r)}{\cos(\phi_r)}\right) \tag{5}$$

where n can be solved for numerically by minimizing this equation.

5.2 Methods

5.2.1 Materials

- Same as Section 3.2.1
- Plate with refraction index n of known or measured thickness d

5.2.2 Experimental Set Up

- Follow the same set up as the two previous experiments.
- Place the plate with the n we want to measure normal to the first mirror. Make sure you are able to swivel the plate.

5.2.3 Experimental Procedure

- Take a picture of the interference pattern when the plate is parallel to the normal mirror.
- Slowly move the plate to an angle ϕ_i and measure the angle. Record the resulting interference pattern.
- Count the interference rings, then solve Equation 5 for n.

5.3 Analysis and Interpretation

Table of our measurements for ϕ_i and n and analysis of interference pattern. Possibly a plot showing the minimization of Equation 5 to arrive at the answer for n.

Uncertainty from measuring the angle and the thickness of the plate.

6 Conclusion

In this section we will report:

- the wavelength of the HeNe laser
- the wavelength of the Thallium spectral line

- the separation between the yellow Mercury doublet lines
- the index of refraction of our transparent material
- .... and their associated uncertainties

We will recap the significance of these measurements and re-iterate the utility of Michelson interferometry for making precise measurements such as these, and other uses it has in physics.

References

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